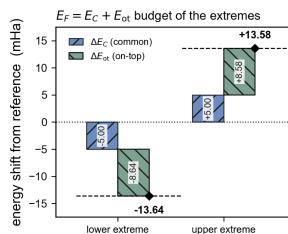
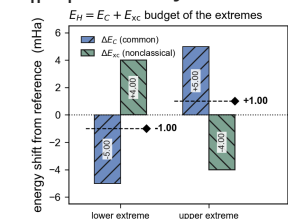
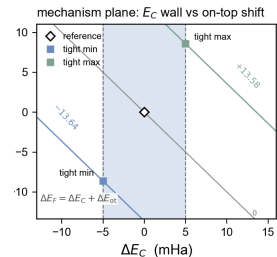


a $E_H - E_F$ sensitivity mismatch



same small 2-RDM moves ($\|\Delta D_2\|_F \leq 1.01\%$):
 $|\Delta E_H| \leq 1$ mHa vs $|\Delta E_F| \approx 13.6$ mHa



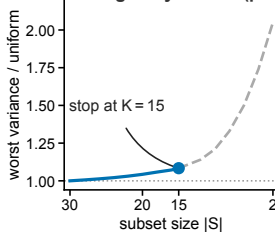
b frame selection

pilot: every frame $\times$ 500 shots



pool: 30 Haar O(8) frames

backward-greedy screen (pilot-only)



keep $K = 15$, equal shots



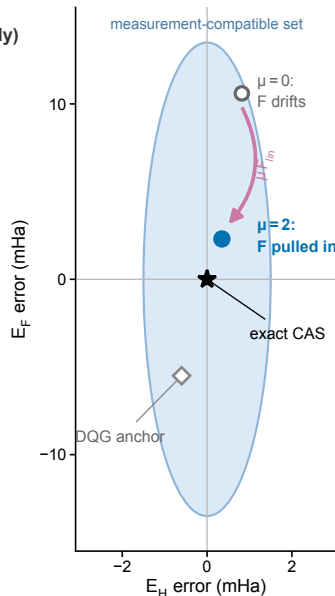
($B - 15,000$)/15 new shots per kept frame
 kept pilots reused, dropped ones discarded

c $\mu = 2$ ftPBE regularization

band SDP on the 15 selected frames

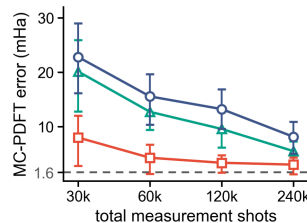
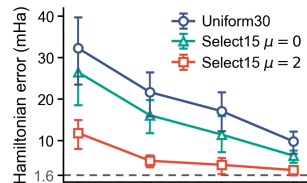
$$\min_{D, \gamma, E^+} H + \mu F_{\text{lin}} + \lambda \text{Tr}(E^+ + E^-)$$

s.t. DQG 2-positivity, spin, contraction,
 row band $\tilde{\gamma} \in A \text{vec}(D \pm \Delta)$



d effect on H and F errors

N_2 , $R = 1.10$ Å



N_2 dissociation, 120k shots

